Quizly — Product Requirements Document

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1. Executive Summary

Quizly is a web-based assessment platform built for businesses, educational institutions, and HR teams that need to create, distribute, and proctor online examinations at scale. The platform handles the full lifecycle of an assessment — from writing questions and inviting candidates, through to test-taking with fraud detection, and finally to scoring, analytics, and reporting.

The product targets organisations that currently rely on manual testing processes, paper-based exams, or generic survey tools that lack the rigour needed for high-stakes evaluations.

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2. Problem Statement

Organisations across industries — oil and gas, healthcare, engineering, finance, technology — regularly need to evaluate candidates, employees, or students through structured assessments. Existing solutions tend to fall into two categories:

- **Generic survey tools** (Google Forms, Typeform) that lack proctoring, fraud detection, timed sessions, and proper scoring.
- **Enterprise exam platforms** (Mercer Mettl, HackerRank) that carry high per-seat costs and lengthy onboarding processes.

There is a gap in the market for a mid-tier product that combines the ease of setup of a form builder with the rigour of an enterprise exam platform — specifically one that supports a wide range of question types (including visual and psychometric formats), offers built-in proctoring, and provides actionable analytics.

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3. Product Vision

Deliver a self-service assessment platform where any business administrator can go from zero to a published, proctored exam in under 15 minutes — without needing a technical background or a dedicated vendor onboarding process.

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4. Target Audience

Segment	Use Case
HR & Recruitment Teams	Pre-employment screenings, aptitude tests, technical evaluations
Corporate L&D	Compliance certifications, safety exams, periodic knowledge checks
Educational Institutions	Course assessments, entrance exams, practice tests
Consulting & Training Firms	Client-facing certifications, post-training evaluations
Industry Safety Bodies	Petroleum safety, electrical safety, healthcare compliance

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5. User Roles

The platform supports three distinct user roles:

5.1 Business Administrator

The primary paying user. Registers an account, subscribes to a plan, and manages all assessments and candidates within their organisation.

Key responsibilities:

- Create and configure assessments
- Add questions (manually, via CSV import, or from the template library)
- Invite and manage candidates
- Publish and schedule assessments
- Review results, scores, and analytics
- Export reports

5.2 Candidate (Test-Taker)

An individual invited to take an assessment. Does not need to create an account — they access the test via a unique invitation link or a shared access code.

Key responsibilities:

- Open the invitation link
- Confirm identity and start the session
- Answer questions within the time limit
- Submit the assessment

5.3 Super Administrator

An internal platform operator with full visibility across all users, assessments, payments, and system settings.

Key responsibilities:

- Monitor platform health and usage
- Manage subscription plans and pricing
- Enable or disable features via feature flags
- View revenue reports and user activity
- Manage assessment templates available to all users

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6. Feature Requirements

6.1 Authentication & Account Management

Feature	Description
Email & Password Registration	Business admins register with name, email, password, and optional company name. Passwords must meet complexity requirements (mixed case, numbers, symbols, 8+ characters).
OTP Email Verification	A 6-digit code is sent to the registered email. The code expires after 10 minutes. Accounts remain unverified until the code is confirmed.
Login	Email and password-based login with brute-force protection (5 attempts per minute). Failed logins return a generic message to prevent user enumeration.

Feature	Description
Password Reset	Token-based password reset flow via email. Reset links expire after use.
Account Deactivation	Super administrators can deactivate a business admin account, which immediately revokes all active sessions.
Session Management	Each login produces a new session token. Previous tokens are revoked on login to enforce single-session behaviour.

6.2 Assessment Creation & Management

Feature	Description
Create Assessment	Define title, description, duration (5–480 minutes), passing score (0–100%), start and end dates.
Assessment Options	Toggle question shuffling, option shuffling, back-navigation, result visibility to candidates, proctoring, fullscreen enforcement, webcam recording, and auto-end on tab switch.
Edit Assessment	Draft assessments can be freely edited. Published assessments only allow changes to scheduling and display settings.
Duplicate Assessment	Clone an existing assessment with all its questions and options into a new draft.
Delete Assessment	Only assessments in draft or cancelled status can be deleted.
Publish Assessment	Publishing validates that the assessment has at least one question and one candidate, then queues invitation emails and generates a public access code.
Assessment Status Lifecycle	Draft → Scheduled (if start date is in the future) → Active → Completed.

6.3 Question Types

The platform supports 22 question types, grouped into the following categories:

Standard Types

- Single Choice — One correct answer from multiple options
- Multiple Choice — One or more correct answers
- True/False — Binary statement evaluation
- Fill in the Blank — Free text answer matched against expected values

- Numeric — Numerical answer with optional tolerance
- Text Input — Open-ended free text response

Advanced Types

- Ordering — Drag items into the correct sequence
- Matching — Pair items from two columns
- Code Snippet — Write code in response to a prompt

Psychometric & Reasoning Types

- Sequence Pattern — Identify the next item in a numerical or symbolic sequence
- Matrix Pattern — Identify the missing element in a matrix grid (Raven's Matrices style)
- Odd One Out — Pick the item that does not belong
- Analogy — Complete an A:B :: C:? relationship
- Mental Maths — Solve arithmetic word problems
- Word Problem — Solve applied reasoning problems
- Spatial Rotation — Identify a rotated or transformed shape
- Pattern Recognition — Identify rules in visual patterns

Interactive Types

- Shape Puzzle — Drag-and-drop shape fitting (similar to Duolingo-style exercises)
- Drag & Drop Sort — Drag items into position
- Hotspot — Select a region on an image
- Likert Scale — Rate agreement on a scale

Visual Types

- SVG Visual Patterns — Visually rendered geometric sequences and matrices stored as structured data and drawn on screen

6.4 Question Management

Feature	Description
Add Questions	Add one at a time with text, type, point value, options, and expected answer.
Edit & Delete Questions	Modify or remove any question before the assessment is published.
Reorder Questions	Manually set the display order of questions.

Feature	Description
CSV Import	Upload a CSV file to bulk-import questions.
Batch Delete	Select and remove multiple questions at once.
Template Import	Browse the platform's template library and selectively import questions into an existing assessment.

6.5 Candidate Management

Feature	Description
Add Candidates	Add individually by email (with optional name), in bulk by pasting a list, or via CSV upload (up to 1,000 per batch).
Edit Candidate	Update email or name for candidates who have not yet started.
Remove Candidate	Remove candidates who have not yet started the test.
Batch Delete	Select and remove multiple candidates at once.
Send Invitations	Queue email invitations in batches of 50 (with staggered delivery to avoid rate limits).
Resend Invitation	Resend the invitation email to a specific candidate.
Batch Resend	Resend invitations to multiple selected candidates at once.
Previous Candidates	Re-invite candidates from other assessments without re-entering their details.
All Candidates View	A searchable, filterable directory of every candidate across all assessments.
Public Access Code	After publishing, a shareable link is generated. Anyone with the link can register and take the test without a prior invitation.

6.6 Test-Taking Experience

Feature	Description
Token Validation	When a candidate opens their link, the system verifies the token, checks that the assessment is active, and confirms they have not already completed it.
Session Start	Candidate confirms their name, the system runs fraud checks, creates a session with a timestamp, and captures device fingerprints.

Feature	Description
Question Display	Questions are rendered one at a time (or with a navigation panel, depending on settings). All 22 question types have dedicated input interfaces.
Answer Saving	Each answer is saved immediately and idempotently. Candidates can return to previous questions if back-navigation is enabled.
Timer	A countdown timer is always visible. When time expires, answers are auto-submitted.
Session Resumption	If a candidate loses connection and returns, their existing session is resumed from where they left off.
Submission	Candidate manually submits, or the system auto-submits on time expiry. Scores are calculated immediately.
Result Display	If the assessment creator has enabled result visibility, the candidate sees their score, pass/fail status, and breakdown.

6.7 Proctoring & Fraud Detection

Feature	Description
Fullscreen Enforcement	The test can require fullscreen mode. Exiting fullscreen is logged and counted.
Tab Switch Detection	Switching away from the test tab is detected and recorded. Optionally, the test auto-ends on tab switch.
Webcam Recording	When enabled, the candidate's webcam feed is recorded and uploaded to cloud storage. The recording is linked to their session and available for review.
Device Fingerprinting	The system captures canvas, WebGL, and screen resolution fingerprints to flag repeat attempts from the same device.
IP Address Logging	Each session records the candidate's IP address. Multiple sessions from the same IP are flagged.
Exact Email Block	A candidate who has already completed the assessment with a given email is blocked from retaking it.
Fuzzy Email Detection	Similar email addresses (based on string similarity) are flagged for review but not blocked.
Name Similarity Detection	Similar names across sessions are flagged for review but not blocked.

Feature	Description
Fraud Attempt Logging	All fraud detection events are logged with full context for later review.

6.8 Analytics & Reporting

For Business Administrators:

Metric	Description
Dashboard Overview	Total assessments, total candidates, completion rate, average score, weekly trends, and month-over-month change.
Activity Feed	Live-updating list of recent events (completions, new invitations).
Assessment Analytics	Per-assessment breakdown: response rate, pass/fail ratio, average score, average completion time, score histogram (0–20, 21–40, 41–60, 61–80, 81–100).
Question Analysis	Success rate for each question, identifying which questions are easiest and hardest.
Completion Funnel	Invited → Started → Completed → Passed.
Top Scorers	Ranked list of highest-performing candidates.
Time Distribution	How long candidates took, bucketed into time ranges.
Session Detail View	For each candidate: score breakdown, answer-by-answer review (with correct answers shown), time spent, proctoring flags (tab switches, fullscreen exits), and webcam recording playback.

For Super Administrators:

Metric	Description
Platform Dashboard	Total users, new registrations this month, active subscriptions, total and monthly revenue.
Revenue Analytics	Total lifetime revenue, monthly revenue trend (12 months), revenue by billing cycle, subscription conversion rate.
Reports	Date-filtered reports with period-over-period comparison, revenue trends, user growth, and downloadable transaction history.

Metric	Description
PDF Export	Generate and download a formatted PDF report of platform activity for any date range.

6.9 Payments & Subscriptions

Feature	Description
Subscription Plans	The platform offers tiered subscription plans with monthly billing. Plans are managed by the super administrator. Each plan lists its price and included features.
Annual Billing	Plans support an annual billing option with a configurable discount percentage.
Payment Processing	Payments are processed through Flutterwave. The business admin is redirected to a hosted checkout page to complete payment.
Payment Verification	Payments are verified through two independent paths: a redirect callback (after the user returns) and a server-to-server webhook (for reliability). Both paths are idempotent.
Subscription Activation	On successful payment, the user's subscription is activated for one month (or one year for annual billing). A confirmation email is sent.
Subscription Gating	Most platform features (creating assessments, adding questions, managing candidates) require an active subscription. Template browsing and account management are available to all registered users.

6.10 Email System

The platform sends the following transactional emails:

Email	Trigger
OTP Verification	When a business admin registers or requests a new verification code.
Assessment Invitation	When an assessment is published or a candidate is manually invited/re-invited.
Assessment Result (to Admin)	When a candidate submits their test. Contains the candidate's score and session details.
Candidate Result	(Optional, controlled by feature flag) When a candidate submits their test. Contains their answers and score.

Email	Trigger
Subscription Confirmation	When a payment is successfully processed.
Contact Form	When a visitor submits the contact form on the public website.

All emails are processed through a background queue to avoid blocking user interactions.

6.11 Assessment Templates

The platform ships with 14 pre-built assessment templates spanning multiple industries and skill areas. These templates serve both as a quick-start tool for new users and as a demonstration of the platform's question type variety.

Available Templates:

1. Petroleum Operations Safety Assessment
2. Software Engineering Technical Assessment
3. Pattern Recognition & Cognitive Assessment
4. Electrical Engineering Fundamentals
5. Mechanical Engineering Assessment
6. Healthcare & Nursing Assessment
7. General Knowledge & Aptitude Test
8. Cybersecurity Fundamentals Assessment
9. Project Management Professional Assessment
10. Data Science & Analytics Assessment
11. Civil Engineering Assessment
12. Accounting & Finance Assessment
13. Visual Abstract Reasoning Assessment
14. Shape Puzzle & Spatial Reasoning Assessment

Templates contain 6–12 questions each using a mix of standard, advanced, and interactive types. Business administrators can clone an entire template or selectively import individual questions from any template into their own assessments.

6.12 Feature Flags

The platform's feature set is controlled by a configurable flag system, allowing the super administrator to enable or disable functionality without redeployment. The following capabilities are individually toggleable:

- Payment processing (global kill switch)
- Flutterwave provider
- Proctoring features
- Webcam recording
- Auto-end on tab switch
- Advanced question types (psychometric and interactive)
- Visual pattern builder
- Question navigation panel
- Email notifications
- Candidate answer reports
- Google OAuth
- User registration
- Maintenance mode
- Analytics dashboard

6.13 Super Admin Panel

Feature	Description
User Management	Search, filter, view, edit, and enable/disable business admin accounts. View each user's assessment count, payment history, and subscription status.
Subscription Plan Management	Create, edit, and delete subscription plans (with pricing, annual discounts, and feature lists). Plans with active subscribers cannot be deleted.
Assessment Oversight	Browse all assessments across all users. View full details, questions, candidates, and results. Delete assessments or promote them to system templates.
Template Administration	Create new assessment templates, add questions to them, and manage the template library.
Revenue Dashboard	View total revenue, monthly trends, billing cycle breakdown, and recent payment history.
Reports	Generate date-filtered platform reports with comparison to the previous period. Export as PDF.
Settings	Manage platform-wide settings with typed key-value pairs (string, integer, float, boolean, JSON).
Feature Flags	Toggle platform features on or off with descriptions and categories.

6.14 Real-Time Updates

The platform uses WebSocket broadcasting to push live updates to connected clients. The following events are broadcast:

- Assessment created, updated, published, or deleted
- Candidate added, removed, or invited
- Test session completed (triggers dashboard refresh with new score data)

Business administrators see their dashboard and candidate lists update in real time without needing to refresh the page.

6.15 Public Website

The platform includes a public-facing website with:

- **Landing Page** — Product overview, feature highlights, and call-to-action for registration.
- **About Page** — Information about the platform and team.
- **Contact Page** — A form that sends an email to the support team.
- **Pricing Page** — Displays available subscription plans with feature comparisons.

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7. Non-Functional Requirements

Requirement	Detail
Rate Limiting	All API endpoints are rate-limited by tier (auth, test-taking, general API). Login is limited to 5 attempts per minute. OTP requests are limited to 3 per 10 minutes.
Input Sanitization	All user-submitted input is sanitized at the middleware level before reaching any controller logic.
Idempotency	Payment initialisation and answer submission are idempotent. Duplicate webhook deliveries are handled gracefully.
Concurrency Safety	Test session creation uses distributed locks to prevent duplicate sessions from race conditions.
Data Integrity	Payment activation uses row-level locking within database transactions to prevent double-processing.
Caching	Admin dashboard data is cached for 30 seconds. Subscription plans are cached for 5 minutes.

Requirement	Detail
Background Processing	Emails, invitation batches, and result notifications are dispatched to a background job queue.
Scalability	All analytics queries use direct SQL aggregation rather than loading full datasets into memory.

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8. Business Model

Quizly follows a subscription-based SaaS model:

- **Free Tier** — Account registration, template browsing, and limited functionality.
- **Paid Tiers** — Full access to assessment creation, candidate management, proctoring, and analytics. Plans are managed and priced by the platform operator.
- **Revenue Stream** — Monthly and annual subscription fees processed through Flutterwave.

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9. Success Metrics

Metric	Target
Time to First Published Assessment	Under 15 minutes from registration
Assessment Completion Rate	Above 70% of invited candidates
Candidate Experience (result delivery)	Score visible within 3 seconds of submission
Real-Time Dashboard Latency	Under 2 seconds for live update propagation
Payment Success Rate	Above 95% for initiated payment flows

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10. Glossary

Term	Definition
Assessment	A timed test or exam containing a set of questions, created by a business administrator.

Term	Definition
Candidate / Test-Taker	An individual who is invited to take an assessment.
Business Administrator	A registered user who creates and manages assessments.
Super Administrator	An internal platform operator with full system access.
Session	A unique test-taking attempt by a candidate for a specific assessment.
Template	A pre-built assessment with questions, available for all users to clone or import from.
Feature Flag	A configuration toggle that controls whether a specific platform feature is enabled.
Proctoring	Monitoring mechanisms (fullscreen, tab switching, webcam) that help ensure test integrity.
Access Code	A public link that allows anyone to register and take a published assessment without needing a prior invitation.

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